

COMMENTARY

Weighing the Options for Limiting Surplus Animals

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The unsustainability of many animal programs managed by zoos and aquariums has brought renewed attention to unresolved questions about various management strategies. Solving the “sustainability crisis” for many species will require housing more adults and producing more offspring than there are existing spaces in accredited zoos and aquariums. Careful reproductive management is central to addressing this challenge, but opinions differ about which management strategies are best for an individual, for a species, for an institution, or for a country or region. The primary options for limiting the number of animals that would be surplus to the population are to prevent reproduction or to euthanize. However, there is much misunderstanding about methods for controlling reproduction, in particular about contraceptives and species differences in their effects. Careful weighing of all the options is called for. Lifetime Reproductive Planning may help increase breeding success through careful reproductive management but cannot eliminate production of surplus animals. Limiting reproduction does not address the problem of animals already in the population. Despite best efforts and planning, consistently hitting target numbers for a population may never be achieved. Increasing capacity provides a temporary patch when targets are exceeded, but is not a long-term solution, since each generation potentially produces even more individuals needing even more space. Welfare considerations should be included in discussions of management euthanasia and its alternatives. Such discussions will be most productive if based on full awareness of the advantages and disadvantages of all the options. *Zoo Biol.* 35:183–186, 2016. © 2016 Wiley Periodicals, Inc.

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The unsustainability of many animal programs managed by zoos and aquariums has brought renewed attention to unresolved questions about various management strategies. It may at first seem paradoxical that reproductive failure should be linked to issues surrounding surplus animals, that is, too few versus too many individuals. However, they are indeed connected in several ways. In the broadest terms, solving the “sustainability crisis” for many species will require housing more adults and producing more offspring than there are existing spaces in accredited zoos and aquariums. Careful reproductive management is central to addressing this challenge, but opinions differ about which management strategies are best for an individual, for a species, for an institution, or for a country or region. The primary options for limiting the number of animals that would be surplus to the population are to prevent reproduction or to euthanize. However, many factors influence decisions about which approach might be right for a particular animal or situation [Penfold et al., 2014].

In presenting results of keeper and program leader surveys on attitudes about management euthanasia, Powell and Ardaiole (this issue) discuss many of the concerns

surrounding that option. The rationale for management euthanasia among EAZA (European Association of Zoos and Aquariums) institutions is that it allows for the natural expression of sexual and parental behaviors, which enriches the lives of the animals and increases welfare [EAZA Culling Statement, 2015]. Euthanizing neonates or pubertal animals approximates nature, representing two of the times when death rates are high, that is, during the first few weeks and around the age of dispersal; euthanizing at weaning should cause less stress to the mother [see Proceedings of the Rigi Symposium, 2003]. However, misinformation about the effects of contraceptives still circulates and confounds

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discussions of advantages and disadvantages of the other options.

WEIGHING THE OPTIONS: CONTRACEPTION

In AZA Institutions, contraception has long been one of the main tools for mammals to reduce production of surplus offspring, by managing reproduction on a year-by-year basis. There are many types of contraceptive products with different actions. There are also profound species differences in response to those products, making careful consideration important for each application.

The AZA Reproductive Management Center (RMC; formerly AZA Wildlife Contraception Center) has never advocated for contraception use. Rather, the Center's mission has been and continues to be providing accurate, up-to-date information for those assessing options for fertility control or who have already decided to use contraception. The mission of the EAZA Group for Zoo Animal Contraception (EGZAC) is similar (www.egzac.org). Both programs generate evidence-based contraceptive recommendations for the major mammalian taxa managed in zoos and aquariums. Based on reports of use in wildlife and from published literature, RMC Recommendations rank available methods and present cautions by taxon (www.stlzoo.org/contraception), but they do not compare contraception to separation of sexes or to management euthanasia.

Evaluation of contraception options, compared to separation or management euthanasia, should weigh accurate and complete information about the contraceptive choices, as well as possible risks or welfare concerns associated with separation of the sexes. The decision can be a complicated one, requiring consideration not only of species differences but also the individual's gender, age, health, social situation, reproductive history, and genetic value to the population.

Contraceptive Efficacy, Safety, and Reversibility

Following reports of uterine and mammary pathology associated with use of progestin contraceptives such as melengestrol acetate (MGA) implants in felids and canids [Munson, 2006; Moresco et al., 2009], many people erroneously assumed these effects occurred in all species. In fact, progestins are used to prevent or reverse uterine pathology in humans [e.g., Diep et al., 2015; Dinh et al., 2015]. Progestin implants and injections and progestin plus estrogen combination pills have been used safely in non-human as well as human primates for decades [RMC Database; Burkman et al., 2004]. Uterine changes sometimes seen in ungulates (e.g., accumulation of fluid) reverse following discontinuation and have not been associated with effects on fertility [Asa, Unpublished Observations]. Steroid-hormone based products such as these are very effective and reverse quickly.

The challenge has been finding a safe, effective, and reversible alternative for carnivores. The GnRH agonist

implant Suprelorin (deslorelin) has been effective in carnivores and most other species (RMC Contraceptive Database). However, if the initial stimulation phase is not blocked, Suprelorin alone can cause pathology in canids (and probably other carnivores) [Asa et al., 2014]. Fortunately, blocking stimulation is simple [Wright et al., 2001]. The major disadvantage of Suprelorin is unpredictable and sometimes very lengthy reversal times, making it a poor choice for animals that might receive a breeding recommendation in the near future.

Effects of Contraception on Behavior

Although the use of contraceptives is unnatural, their effects on behavior or social organization may not differ substantially from natural reproductive conditions. Use of progestin (synthetic progesterone) products approximates the hormone pattern of pregnancy, which is characterized by sustained progesterone elevation. Thus, behavior changes in the treated individuals should be similar to those seen during pregnancy. For great apes given combination birth-control pills (synthetic estrogen plus progestin), a pill-free or placebo week is usually associated with sexual behavior, much like a monthly estrous period. Natural estrous cycles continue at species-typical intervals during porcine zona pellucida (PZP) vaccine treatment. In contrast, Suprelorin, by blocking production of gonadal steroids, is most similar to castration or ovariectomy, so sexual behavior should be reduced or eliminated.

WEIGHING THE OPTIONS: PERMANENT STERILIZATION

The obvious drawback to permanent sterilization is that reproduction will not be possible if genetic rankings change. However, sterilization might be preferable, on welfare grounds, for older females whose health might be jeopardized by copulation, pregnancy, parturition, or contraceptive treatment. Sterilization may be an appropriate choice even for younger animals of species prone to reproductive pathology, such as African painted dogs with high pyometra rates [Asa et al., 2014]. It also precludes the need for disturbance to administer contraceptives, while allowing animals to remain in social groups. Ovariectomy eliminates sexual behavior, but changes are not as predictable for males, since sexual behavior can be a learned response. Vasectomy or tubal ligation allows continued sexual behavior, but should not be used for carnivores with induced ovulation, since sustained progesterone production following ovulation can cause uterine and mammary pathology [Munson et al., 2005].

WEIGHING THE OPTIONS: SEPARATION

The main alternative to contraception for AZA populations was traditionally separation of males from

females. Physiological effects of separation are not benign. Our study of canids [Asa et al., 2014], in addition to similar reports for cheetahs, elephants, rhinoceros, and bats [Hildebrandt et al., 2000; Agnew et al., 2004; Hermes et al., 2004, 2006; Napier et al., 2009; Crosier et al., 2011; Wachter et al., 2011], demonstrated an association between years non-reproductive (no access to males) and infertility. For canids, the risk extends beyond fertility and includes increased incidence of a pyometra, a potentially fatal uterine infection. Systematic studies have not been conducted on more taxa, but reports to the RMC suggest that primates may be less subject to this “use it or lose it” phenomenon than carnivores.

REPRODUCTIVE MANAGEMENT AND WELFARE

It could be argued that, for most species, separation is even more unnatural than contraception in terms of effects on social groups and behaviors. The potential stress of such separation is likely lower for species that tend to be more solitary, but no species with internal fertilization can be completely solitary. The drive to reproduce is strong across all species, as it must be for a species to survive. Studies with lab animals show that individuals expend considerable effort and endure discomfort and pain to gain access to a sexual partner [López et al., 1999]. Separation of pair-bonded individuals is likely to induce an even stronger stress response. Separation typically requires additional space, and unfortunately that additional space may not be of the same size or quality. With the potential for reducing welfare in many species, separation should not be considered a benign option.

Contraception has been generally seen by the AZA community as providing enhanced welfare, because contraception allows family and social groups to remain together while reproduction is prevented in group members not recommended to breed at that time. However, contraception may influence social dynamics, at minimum because most forms of contraception prevent female ovulatory cycles, so courtship and mating behavior do not occur. In fact, for those products, estrous behavior or mating is a useful signal that the contraceptive might not be working. Prolonged absence of young, achieved by any method, is unnatural. Especially in species with complex societies, such as some primates, the absence of playmates may affect social development in youngsters [Suomi et al., 1976].

The use of contraception has been criticized as unnatural by some, in particular because, regardless of the method, the treated female does not experience pregnancy, giving birth, and maternal behavior. However, most females in AZA animal programs do receive one or more breeding recommendations in their lives; between breeding recommendations they may be contracepted. Contraception, by definition, is expected to be reversible. In contrast, sterilization refers to methods intended to be permanent. Thus, contraception is chosen either for females that have

already reproduced, and so have experienced giving birth and caring for young, or that are expected to receive a breeding recommendation in the future. The typical use of contraception does not prevent females from ever reproducing but prevents them from reproducing as often as they otherwise might. The strategy is to space reproductive events to meet population goals, which change from year to year as others in the population are born or die.

CULTURAL ATTITUDES ABOUT LIMITING REPRODUCTION

Attitudes about managing reproduction in pet cats and dogs vary by country or region. In the United States, the common, accepted practice for pet dogs and cats is to spay (ovariohysterectomize) females and neuter (castrate) males. In contrast, in some European and Scandinavian countries, such surgical interventions are not considered justified for fertility control. A study from Sweden on rates of pyometra in unsprayed female dogs documented high rates of uterine disease attributed to years without reproducing [Hagman, 2004]. Nevertheless, most people in the United States accept that most pets are not given the opportunity to reproduce and raise young. Taking the perspective on cultural attitudes a step further, it is not uncommon in the United States for women to choose not to have children, and women who do choose to have children commonly limit the number of children. Studies from around the world have reported that if contraception is available, many (most?) women choose to limit the size of their families [e.g., Gates Foundation, 2015; WHO, 2015]. These observations demonstrate that there is not a universal drive in human females to reproduce repeatedly, on every fertile cycle, although that would be “natural.”

Pregnancy and lactation are energetically very demanding. For captive wildlife, euthanizing offspring at birth eliminates lactational stress on the mother but prevents expression of maternal care. Even if offspring are left with the mother until weaning, the increased energetic demands are likely met by the diet provided by caretakers, making the situation not directly comparable to poor women in developing countries, who might be without adequate nutrition. However, beyond nutritional stress, pregnancy and parturition do carry direct risks to the mother's health, especially if pregnancies are closely spaced [e.g., human data: Kost et al., 1991]. A growing body of literature also indicates that longevity is negatively affected by reproductive rate [e.g., human: Westendorp and Kirkwood, 1998], perhaps via immune system effects [Deerenberg et al., 1997]. The relevance for females in captive breeding programs is that allowing repeated pregnancies without intervening periods of rest and recovery, even if natural, may not be providing the best possible welfare. Although zoos in general do strive to provide natural conditions, being predated or parasitized are also natural, yet are not deemed acceptable by most accredited zoos and aquariums. That is, being natural

may not be in and of itself sufficient justification; natural conditions may not provide the best welfare.

LIFETIME REPRODUCTIVE PLANNING

One approach to more effective reproductive management is Lifetime Reproductive Planning. Fertility would be established by breeding soon after puberty and maintained by breeding every 2–3 years. Once a female's target contribution to the population is reached, she would be treated with a long-acting contraceptive or permanently sterilized. Computer simulations are being developed by the RMC in collaboration with Dr. K. Traylor-Holzer to evaluate various scenarios for individual females and their effects on gene diversity and demography. The objective is to find breeding intervals that maintain fertility but minimize production of surplus young.

CONCLUSIONS

Lifetime Reproductive Planning and contraception will not completely solve the problems of unsustainability or production of surplus animals. Limiting reproduction does not address the problem of animals already in the population. Despite best efforts and planning, consistently hitting target numbers for a population may never be achieved. Increasing capacity provides a temporary patch when targets are exceeded, but is not a long-term solution, since each generation potentially produces even more individuals needing even more space. The fate of some surplus individuals, for example, housed in isolation or crowded into small enclosures, does not provide good welfare. Welfare considerations should be included in discussions of management euthanasia and its alternatives. Such discussions will be most productive if based on full awareness of the advantages and disadvantages of all the options.

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